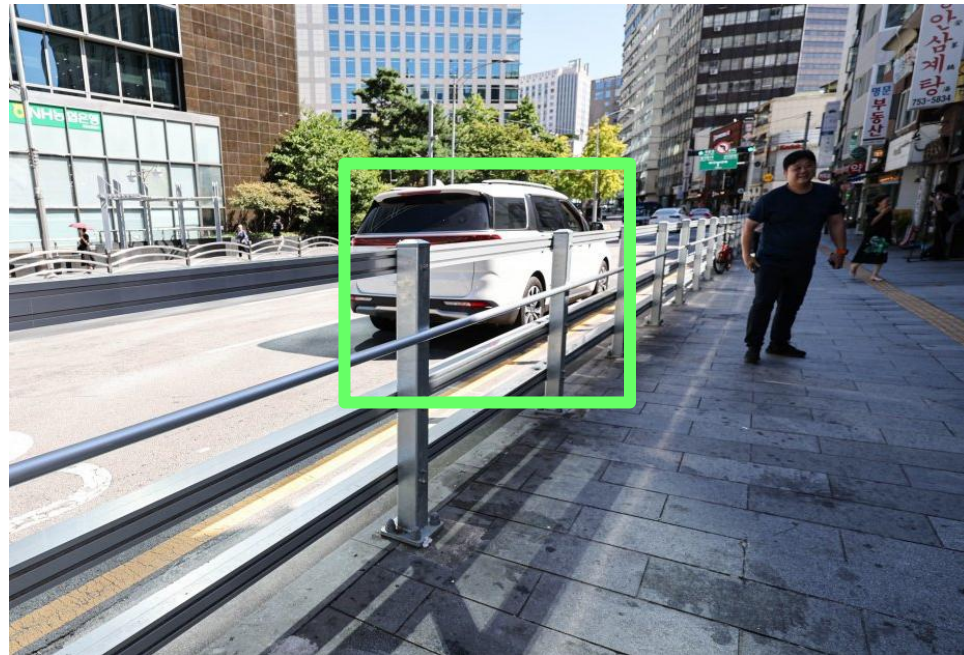


AI기반 ADAS Occlusion 데이터셋 자동 생성 플랫폼

팀명	JOY
팀원	조주영 이재권 이지연 조영진

✓ 프로젝트 개요



Occlusion은 물체가 다른 물체에 의해 가려지는 현상
예: 가드레일, 벽, 구조물 등에 의해 차량이 부분적으로 보이지 않는 상황

이로 인해 ADAS(첨단 운전자 지원 시스템)가 차량을 정확하게 인식하지 못할 수 있음

✓ 프로젝트 개요



nature

Enhancing YOLO for occluded vehicle detection with grouped orthogonal attention and dense object repulsion (2024)

scientific reports

OPEN Enhancing YOLO for occluded vehicle detection with grouped orthogonal attention and dense object repulsion

Jinpeng He, Huaixin Chen[✉], Biyuan Liu, Sijie Luo & Jie Liu

In real-life complex traffic environments, vehicles are often occluded by extraneous background objects and other vehicles, leading to severe degradation of object detector performance. To address this issue, we propose a method named YOLO-OVD (YOLO for occluded vehicle detection) and a dataset for effectively handling vehicle occlusion in various scenarios. To highlight the model attention in unobstructed region of vehicles, we design a novel grouped orthogonal attention (GOA) module to achieve maximum information extraction between channels. We utilize grouping and channel shuffling to address the initialization and computational issues of original orthogonal filters, followed by spatial attention for enhancing spatial features in vehicle-visible regions. We introduce a CloU-based repulsion term into the loss function to augment the network's localization accuracy in scenarios involving densely packed vehicles. Moreover, we explore the effect of the knowledge-based Laplacian Pyramid on the OVD performance, which contributes to fast convergence in training and ensures more detailed and comprehensive feature retention. We conduct extensive experiments on the established occluded vehicle detection dataset, which demonstrates that the proposed YOLO-OVD model significantly outperforms 14 representative object detectors. Notably, it achieves improvements of 4.7% in Precision, 3.6% in AP@0.5, and 1.9% in AP@0.5:0.95 compared to the YOLOv5 baseline.

Keywords Occluded scenarios, Vehicle detection, Grouped orthogonal attention, Laplacian pyramid, Repulsion loss

Vehicle detection is one of the important topics in the field of computer vision, which has a wide range of applications in driverless driving, vehicle counting, and intelligent transportation^{1–3}. Object detector frameworks have made significant strides in accuracy and robustness in recent years, yet obstacles such as occlusion still limit them from reaching the level of human vision⁴. In the context of complex urban traffic, vehicles are often occluded by buildings, trees, pedestrians, and other vehicles, making them difficult to detect due to varying locations, proportions, and types of occlusions. The detection results of different models for two typical scenarios are shown in Fig. 1a to c.

In the field of vehicle detection, occlusion can be divided into two main categories: mutual occlusion among vehicles, known as intra-class occlusion, and occlusion of vehicles by different entities, referred to as inter-class occlusion. In inter-class occlusion scenarios, occlusion of extraneous objects leads to the loss of vehicle features, hinders the expression of key feature information of the inspected vehicle, and makes it difficult for the object detector to extract and learn features of vehicles. In contrast, when occlusion occurs between vehicles, the region of interest (RoI) of the occluded vehicle may contain features of other vehicles, which not only interferes with the accurate localization of the vehicle, but also may cause a high degree of overlap between prediction frames. In the post-processing stage, the detector is very sensitive to the threshold of non-maximum suppression (NMS), and it is difficult to find a threshold t that applies to both sparse and dense objects, making it impossible to handle sparse and highly overlapping objects simultaneously^{5–11}.

To address the issue of occlusion, some methods^{12–14} improve occlusion detection performance by designing part detectors based on the structural features of the occluded target, using a priori knowledge and structural information about the visible parts of the target. The need to train each part leads to an exponential increase

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<https://doi.org/10.1038/s41598-024-70695-x>

nature portfolio

www.nature.com/scientificreports/

$$\delta = \frac{v}{(1 - \text{IoU}) + v}, \quad (11)$$

where b and b^g respectively denote the centroids of the prediction and ground-truth bounding boxes, and then the square of the Euclidean distance is calculated. c denotes the diagonal length of the minimum closure region of the prediction and true frames. In this work, we categorize all vehicles into a single class, thereby transforming it into a binary classification problem. Moreover, the regression loss goal of YOLOv5 is consistent with the L_{box} item target of RepLoss, leading us to maintain YOLOv5's regression loss. Finally, we discard the L_{RepBox} term, which requires significant computational resources, to get our definitive loss function:

$$L = \lambda_1 L_{\text{box}} + \lambda_2 L_{\text{RepLoU}}^{\text{CloU}} + \lambda_3 L_{\text{obj}} + \lambda_4 L_{\text{cls}}, \quad (12)$$

where L_{box} represents the bounding box regression loss, $L_{\text{RepLoU}}^{\text{CloU}}$ represents the repulsion loss, L_{obj} represents the confidence loss, and L_{cls} represents the class loss. The coefficients λ_1 , λ_2 , λ_3 and λ_4 are the equilibrium loss weights, which are set to 0.15, 0.15, 1.0, 0.05 respectively. L_{box} is calculated utilizing CloU loss, while L_{obj} and L_{cls} are computed using binary cross-entropy loss.

Experiments and discussion OVD dataset

One of the current challenges in occluded vehicle detection is the scarcity of datasets of occluded vehicles in authentic traffic environments. Most of the currently available datasets are synthetic, which not only increase the training cost but also may not generalize well to real-world scenarios or adapt to the complexity of the occlusion problem when training models on such datasets⁴. Therefore, to train and evaluate the network we proposed, we establish an occluded vehicle detection dataset based on the UA-DETRAC dataset, named OVD (Occluded Vehicle Detection Dataset). UA-DETRAC stands out as a challenging large-scale dataset for real-world vehicle detection and tracking, which was mainly captured on road overpasses in Beijing and Tianjin and manually annotated with 8250 vehicles and 1.21 million object boxes. However, directly applying UA-DETRAC to our task suffers from the following drawbacks: (1) The UA-DETRAC dataset is a video sequence frame dataset with short inter-frame intervals, resulting in image redundancy; (2) The dataset contains instances of duplicate annotations; (3) A majority of the images lack scenarios with vehicle occlusion; (4) The annotations contain numerous invalid vehicles that are completely occluded; (5) Vehicles occluded by other objects or situated at the image periphery (which can also be considered as occluded) are frequently unlabeled. To address these issues, we employ methods including IoU filtering and manual annotation, resulting in a refined OVD dataset. Specifically, we retain one image every ten frames, discarding those without occlusion objects or where objects are completely obscured. Furthermore, we have meticulously annotated all instances of occluded vehicles within the UA-DETRAC dataset.

Our OVD dataset consists of 8677 images of vehicles in intricate traffic scenarios, with each image abundant in instances of vehicles occluding each other and being occluded by other objects. Fig. 6 displays two typical

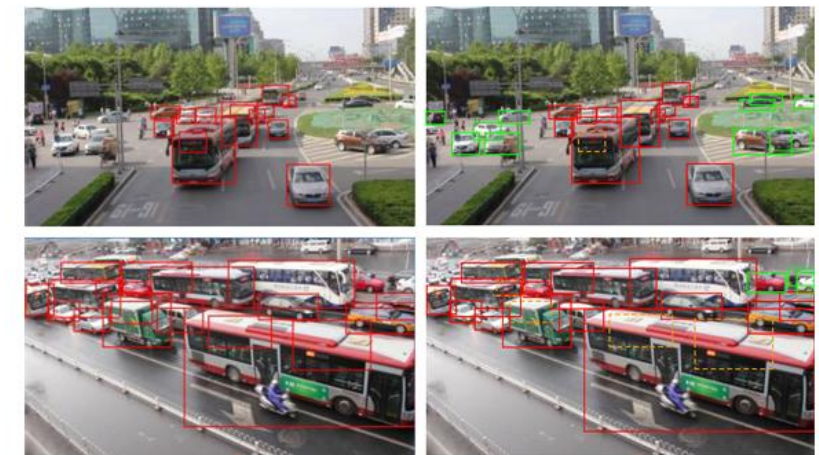


Figure 6. Examples of different scenarios in the OVD dataset (right) compared with the initial dataset (left). The red boxes indicate the original annotations, the green boxes represent the newly annotated vehicles, and the orange dashed boxes denote the discarded annotations.

Scientific Reports | (2024) 14:19650 |

<https://doi.org/10.1038/s41598-024-70695-x>

nature portfolio

✓ 프로젝트 개요

$$\delta = \frac{v}{(1 - \text{IoU}) + v}, \quad (11)$$

where b and b^{gt} respectively denote the centroids of the prediction and ground-truth bounding boxes, and then the square of the Euclidean distance is calculated. c denotes the diagonal length of the minimum closure region

One of the current challenges in occluded vehicle detection is **the scarcity of datasets of occluded vehicles** in authentic traffic environments.

$$L = \lambda_1 L_{box} + \lambda_2 L_{RepGT}^{CIoU} + \lambda_3 L_{obj} + \lambda_4 L_{cls}, \quad (12)$$

where L_{Box} represents the bounding box regression loss, L_{RepGT}^{CIoU} represents the repulsion loss, L_{obj} represents the confidence loss, and L_{cls} represents the class loss. The coefficients $\lambda_1, \lambda_2, \lambda_3$ and λ_4 are the equilibrium loss weights, which are set to 0.15, 0.15, 1.0, 0.05 respectively. L_{Box} is calculated utilizing CIoU loss, while L_{obj} and

“현재 폐색 차량 탐지에서 직면한 과제 중 하나는 실제 교통 환경에서 폐색 차량의 데이터셋이 부족하다는 점입니다.”

Experiments and discussion
OVD dataset

One of the current challenges in occluded vehicle detection is the scarcity of datasets of occluded vehicles in authentic traffic environments. Most of the currently available datasets are synthetic, which not only increase the training cost but also may not generalize well to real-world scenarios or adapt to the complexity of the occlusion problem when training models on such datasets⁴. Therefore, to train and evaluate the network we proposed, we establish an occluded vehicle detection dataset based on the UA-DETRAC dataset, named OVD (Occluded Vehicle Detection Dataset). UA-DETRAC stands out as a challenging large-scale dataset for real-world vehicle detection and tracking, which was mainly captured on road overpasses in Beijing and Tianjin and manually annotated with 8250 vehicles and 1.21 million object boxes. However, directly applying UA-DETRAC to our task suffers from the following drawbacks: (1) The UA-DETRAC dataset is a video sequence frame dataset with short

✓ 프로젝트 목표



주행 영상에서 Occlusion 상황 이미지 추출 자동화

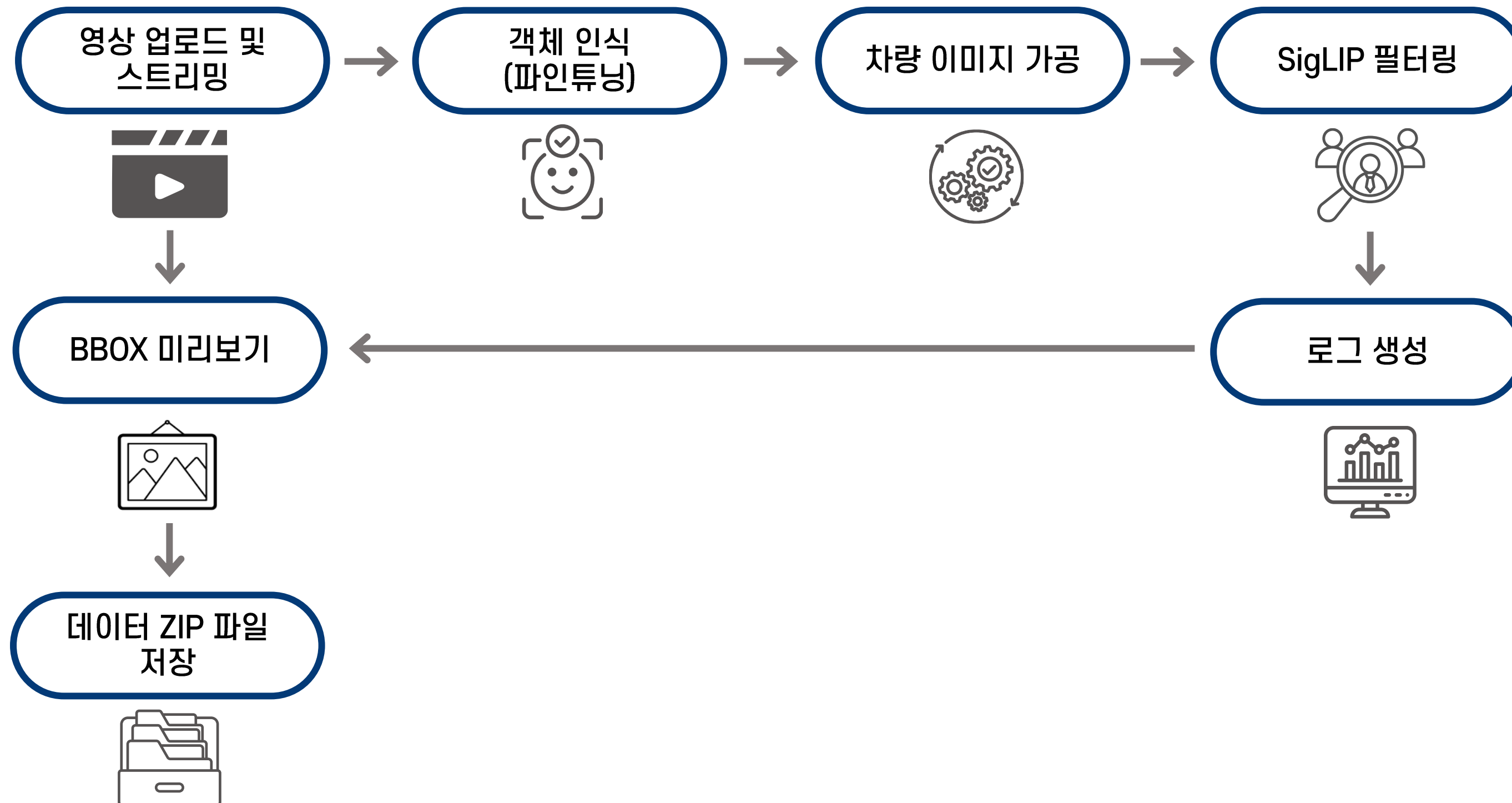
사용자 : ADAS 엔지니어

데이터 수집원 : 주행 영상
플랫폼, 액션캠(DJI)

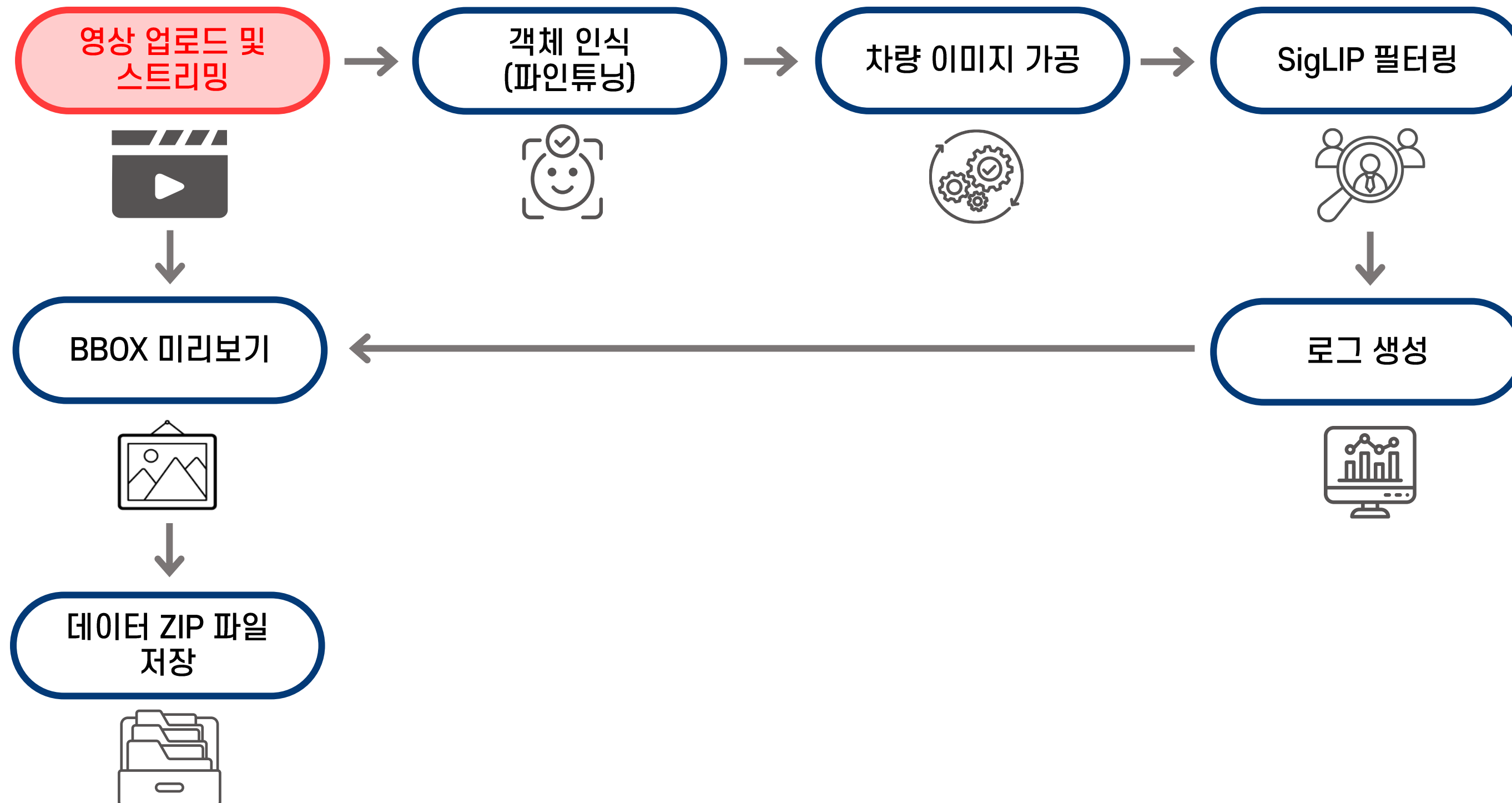
데이터 형식 : JPG

라벨링 형식 : YOLO format

✓ 플로우 차트



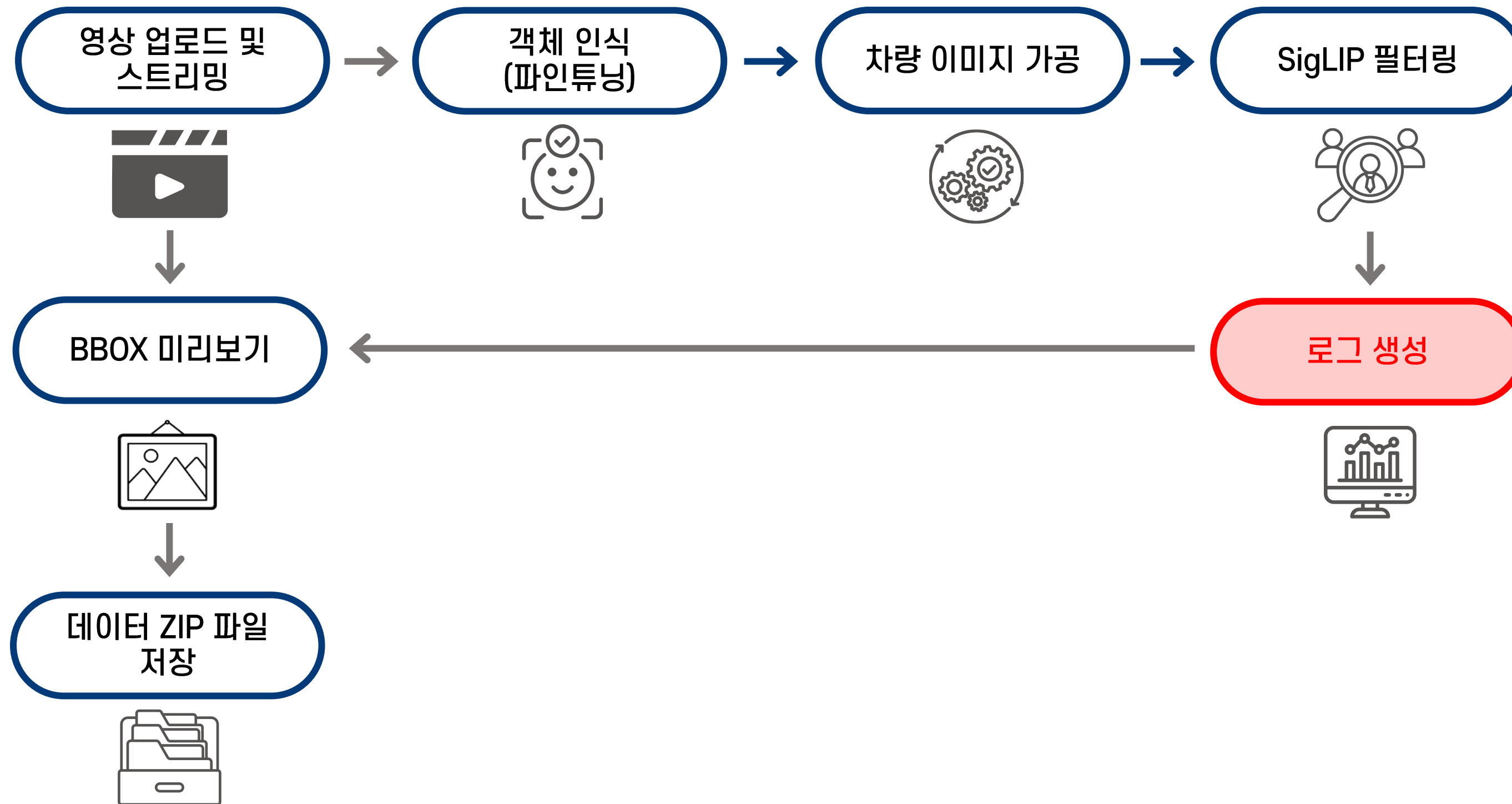
✓ 플로우 차트



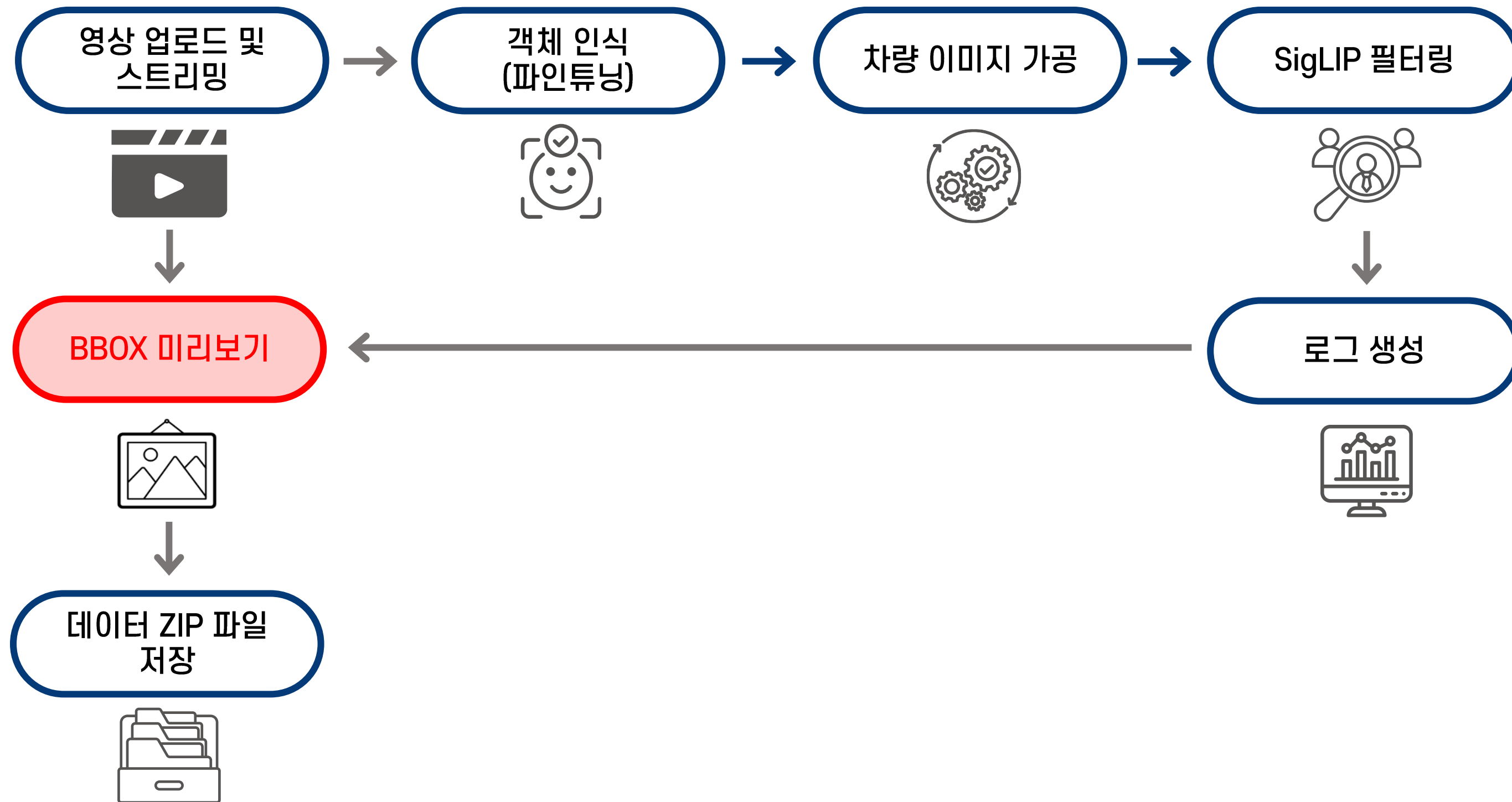
✓ 플로우 차트



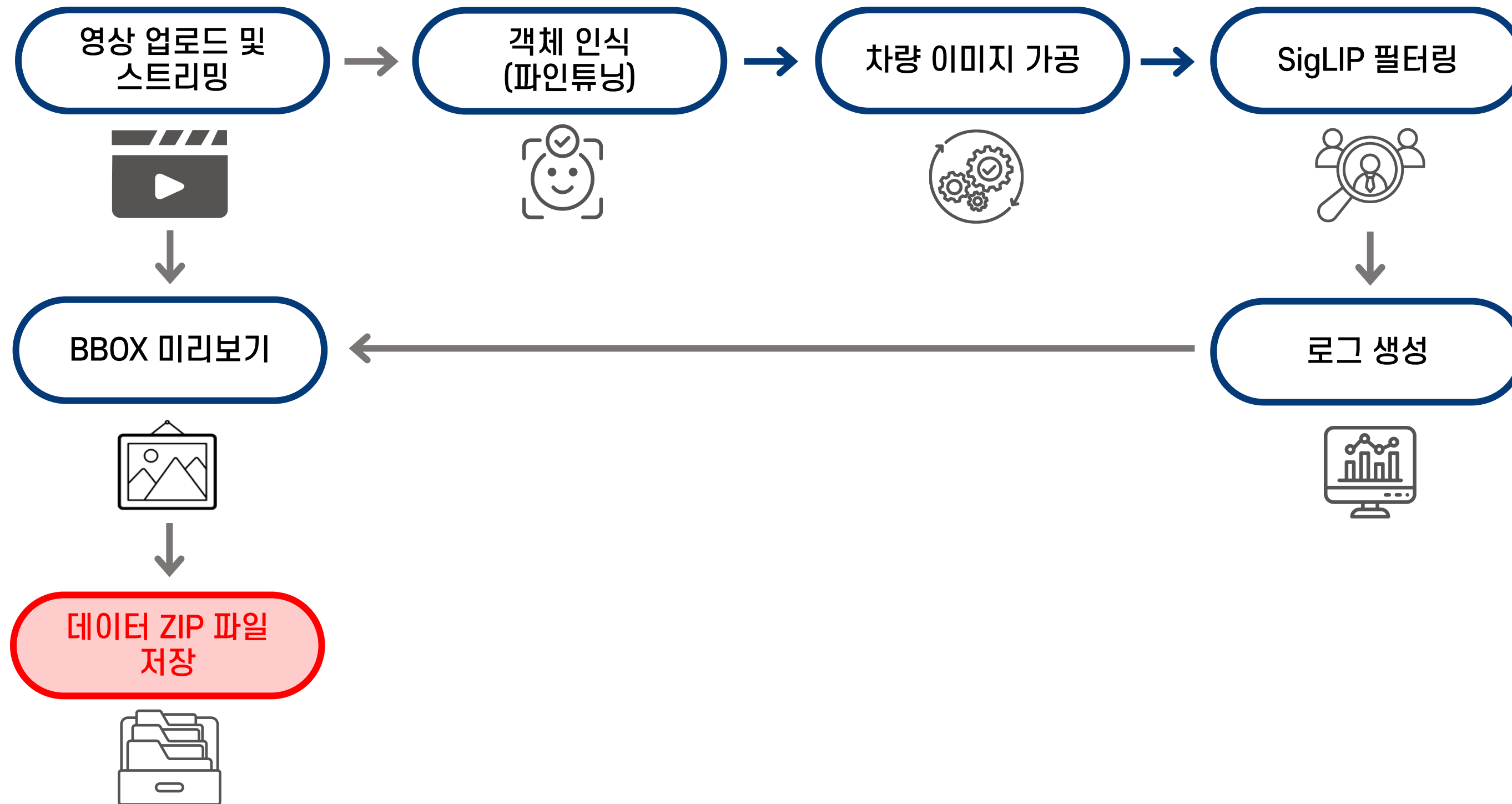
✓ 플로우 차트



✓ 플로우 차트



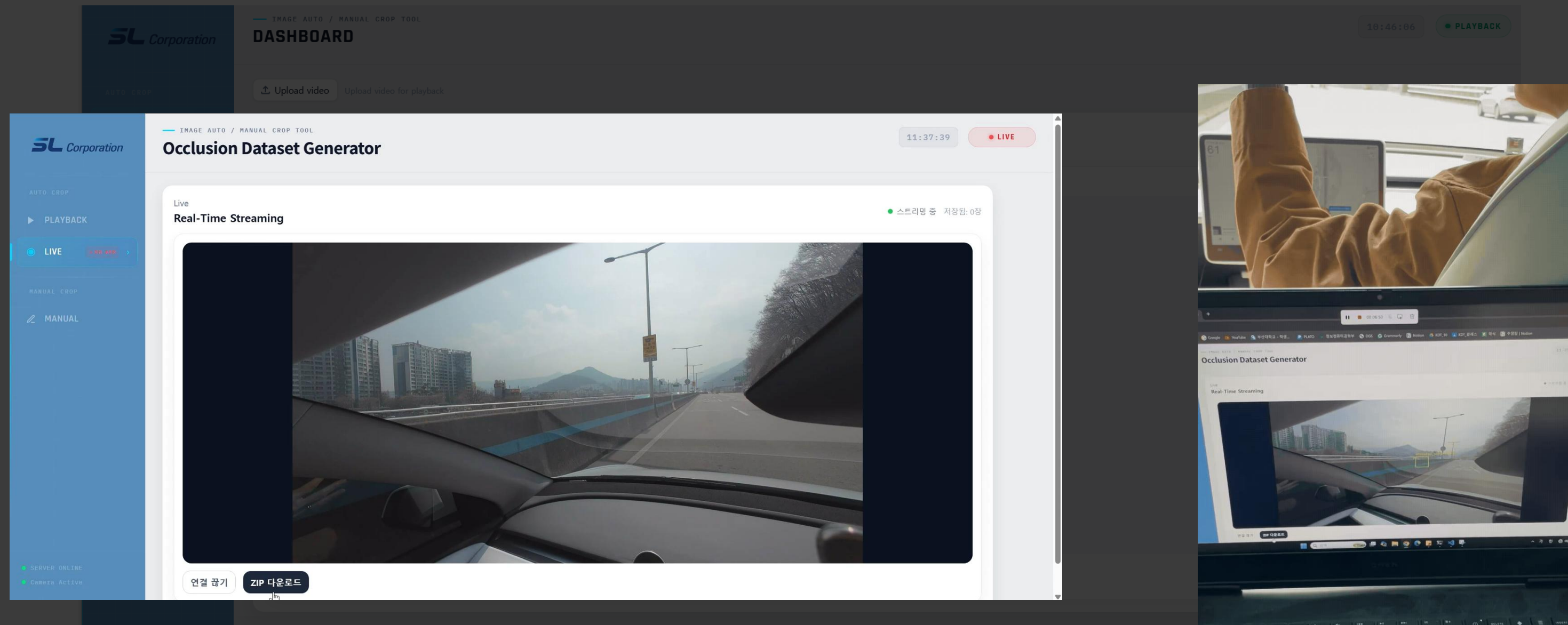
✓ 플로우 차트



✓ 시연 영상 :: Playback

The screenshot displays the SL Corporation DASHBOARD interface. On the left, a sidebar contains navigation options: AUTO CROP, **PLAYBACK** (highlighted), LIVE, MANUAL CROP, and MANUAL. The main content area is titled "DASHBOARD" and features a "Playback" section with the subtitle "Video Playback with AI Detection". A button labeled "Upload Video" with a plus icon is visible, accompanied by the text "동영상을 업로드하세요". The central video player area shows a large blue play button icon and the text "동영상 분석 및 데이터셋 추출" (Video Analysis and Dataset Extraction). Below this, a smaller text block states: "동영상을 업로드하면 AI가 자동으로 분석하고 Occlusion 상황의 데이터셋을 생성할 수 있습니다." (When you upload a video, AI will automatically analyze it and generate a dataset for Occlusion situations). The video player includes a progress bar, seek controls, and volume settings. At the bottom, there is a file upload dialog box with a text input field for "파일 이름(N):", a dropdown menu for "동영상 파일 (*.mpe;*.mpeg;*.o)", and buttons for "열기(O)" (Open) and "취소" (Cancel). The footer of the dashboard includes links for "후보군" (Candidates), "미리보기 (0장) 필터링 결과" (Preview (0 pages) Filtering Results), "미리보기 (0장 중 0장) 추출 로그" (Preview (0 of 0 pages) Extraction Log), and "추출 로그" (Extraction Log).

시연 영상 :: Live



필드 테스트 2026.03.17(Wed) AM11:30
대구 신천대로 경대교 - 두산교 왕복

✓ AI 모델 적용



✓ 객체인식 - 모델 선정

YOLO

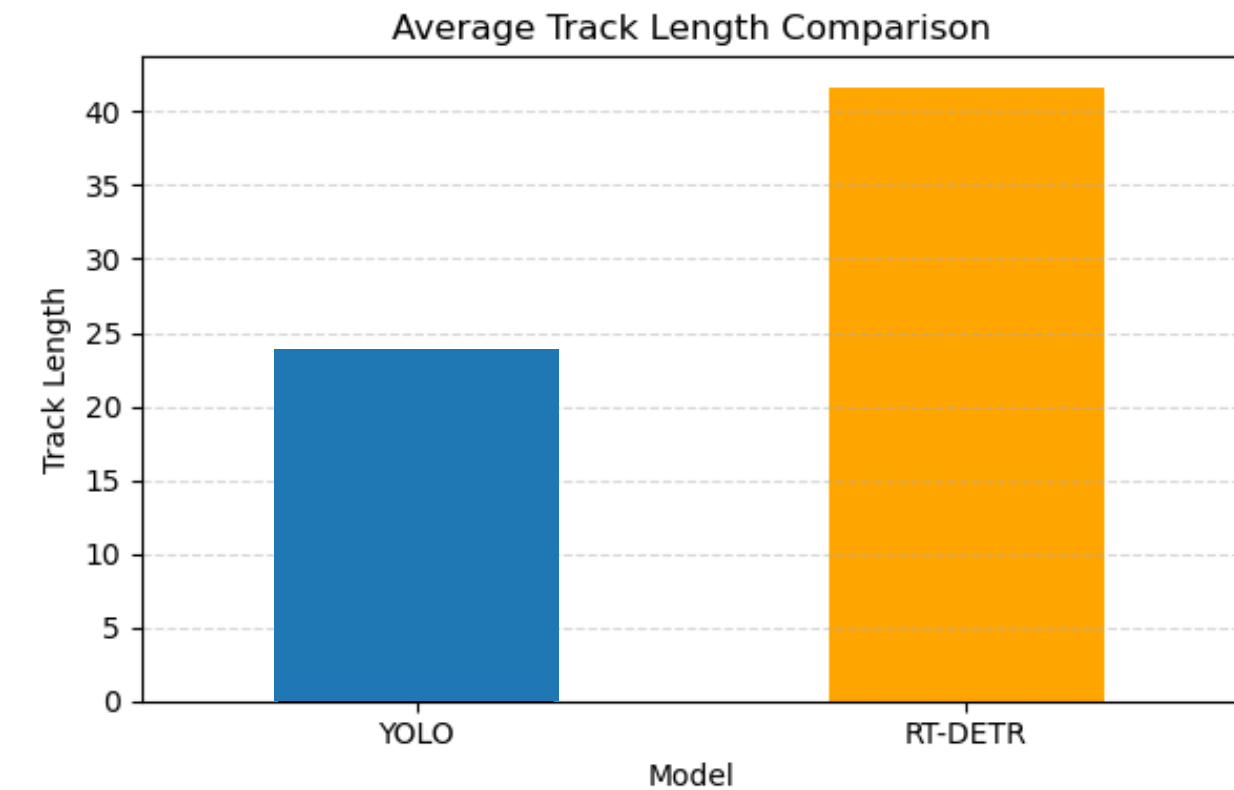
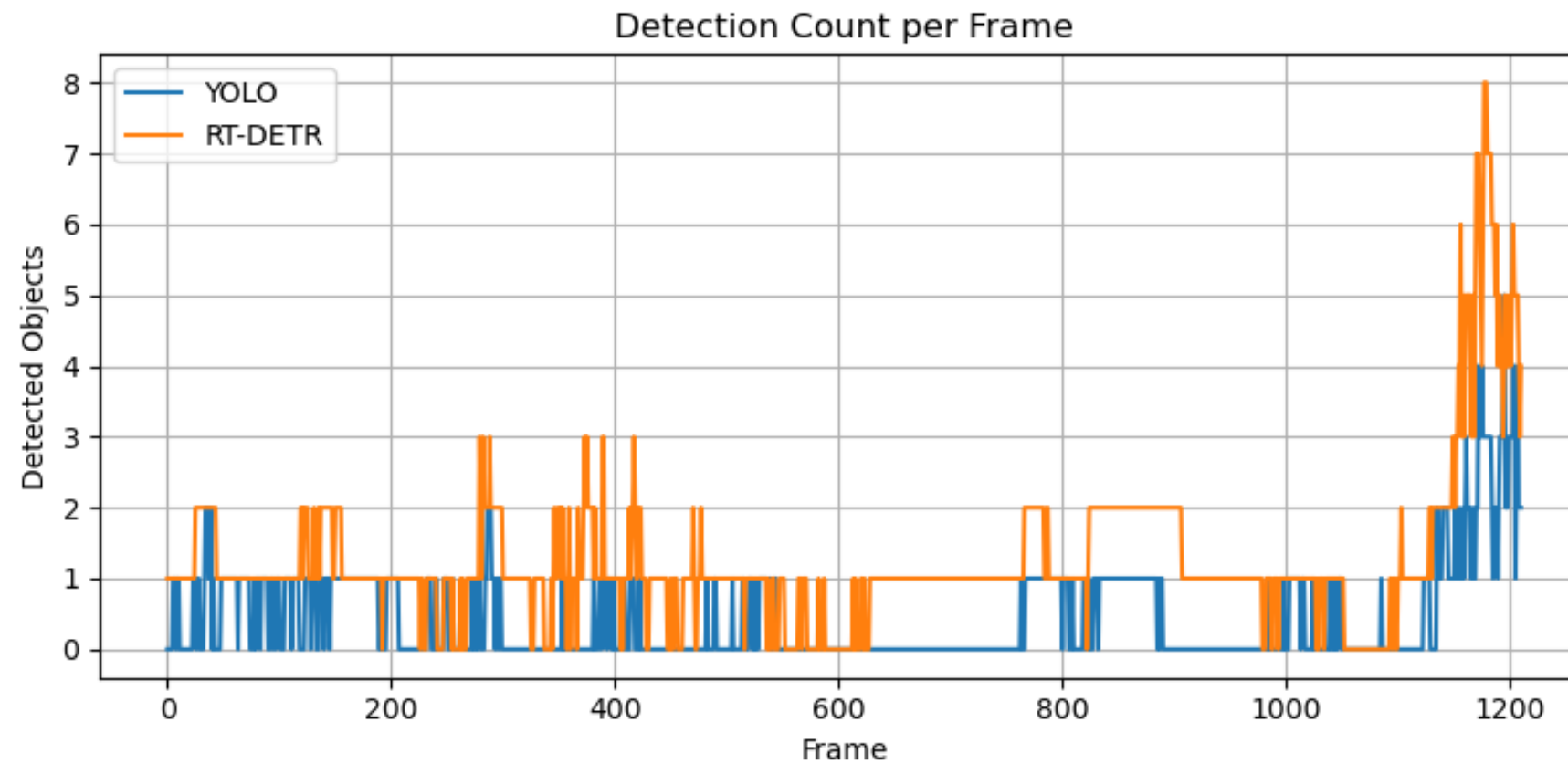
이미지를 격자로 쪼개서 분석

RT-DETR

이미지 전체의 문맥과 관계 파악



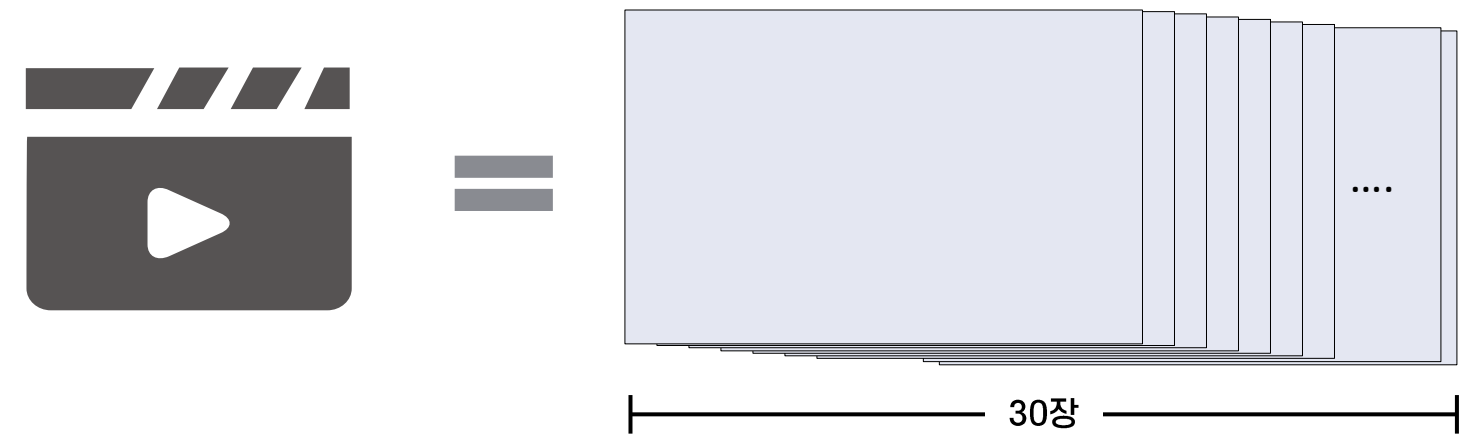
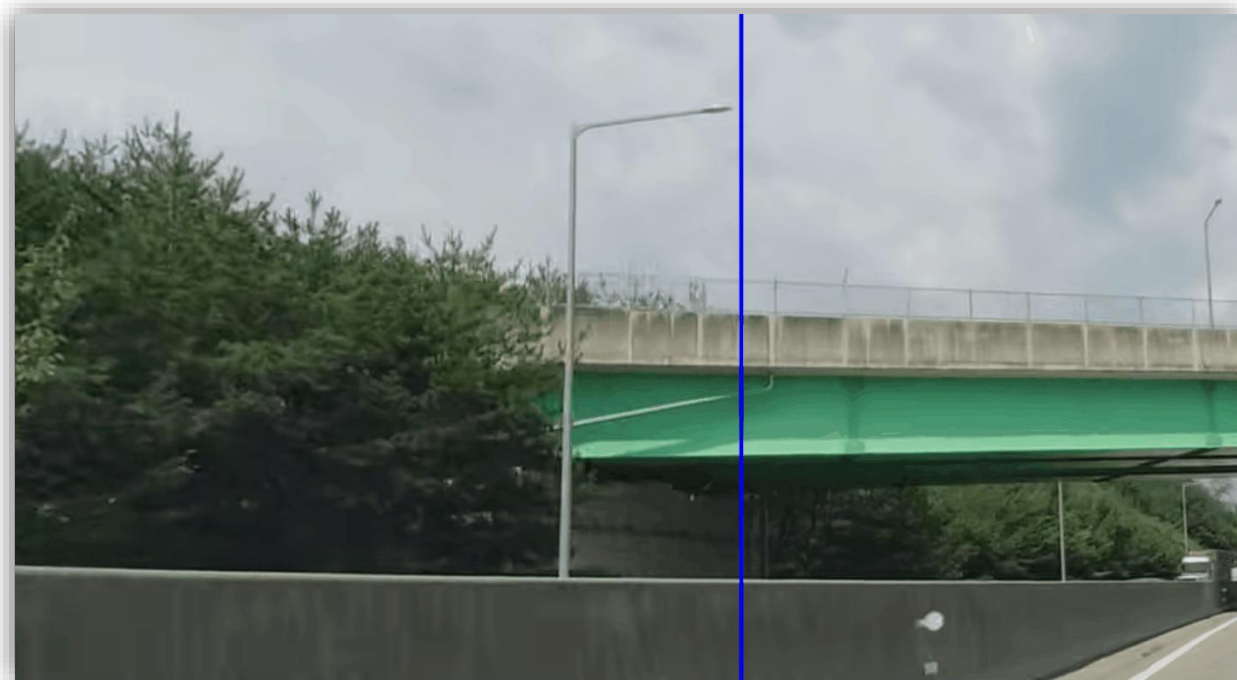
✓ 객체인식 - 모델 선정



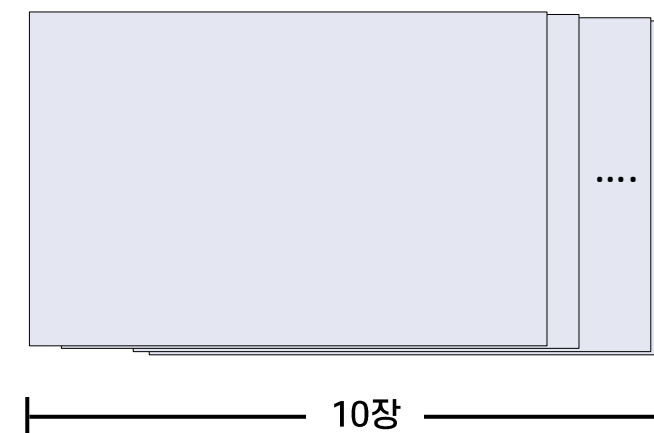
인사이트

- ✓ 모든 프레임 YOLO 대비 RT-DETR가 인식된 객체수가 더 많음.
- ✓ YOLO 대비 RT-DETR의 평균 객체 추적 시간이 길.

✓ 객체인식 - 파인튜닝

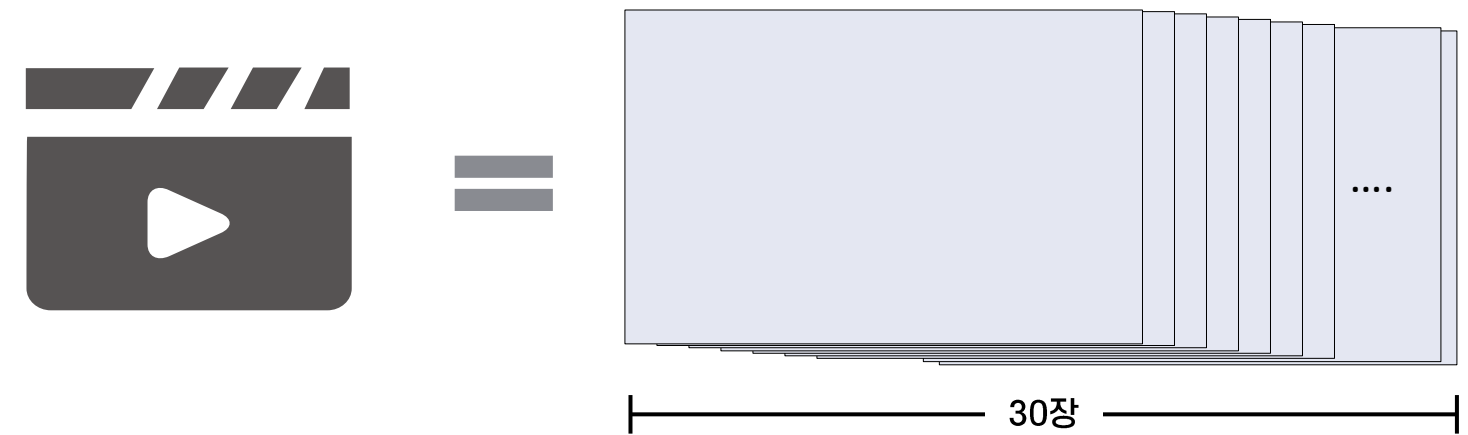
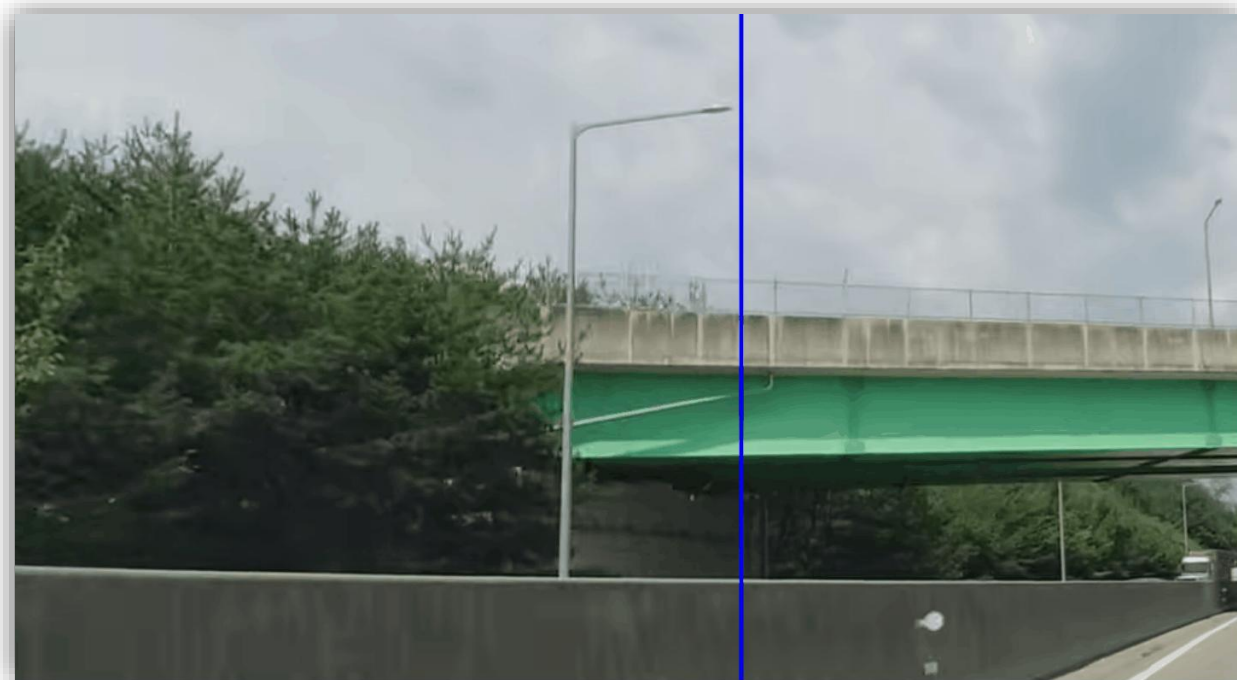


30 프레임 :: 영상 1초에 이미지 30장을 보여주는 것

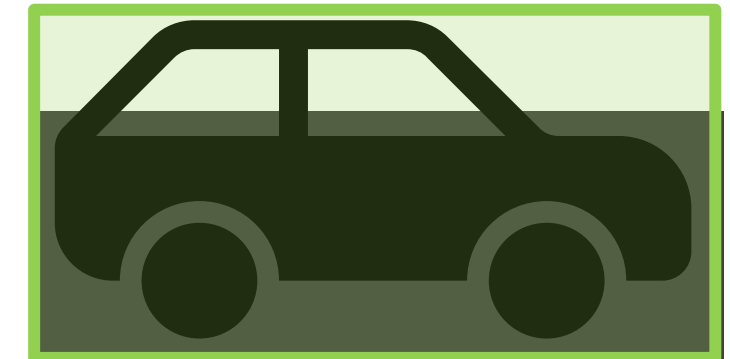
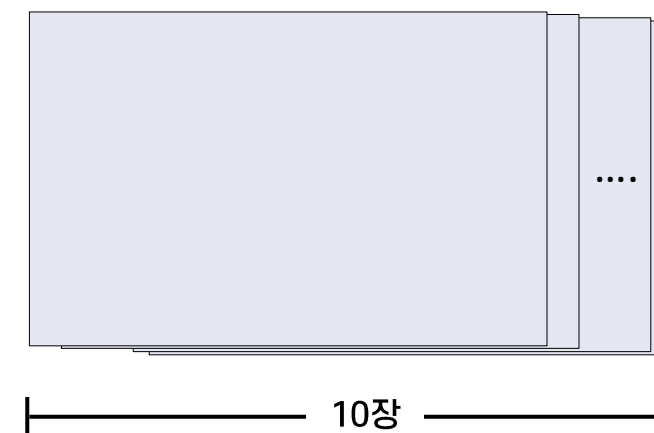


인식된 객체를 10프레임 동안 유지하고
종횡비 ≥ 20 이면 bbox를 아래로 늘려 이미지 저장

✓ 객체인식 - 파인튜닝



30 프레임 :: 영상 1초에 이미지 30장을 보여주는 것



인식된 객체를 10프레임 동안 유지하고
종횡비 ≥ 20 이면 bbox를 아래로 늘려 이미지 저장

✓ 객체인식 - 파인튜닝

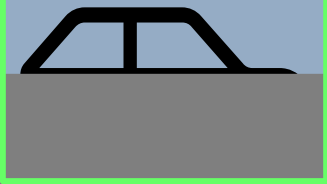
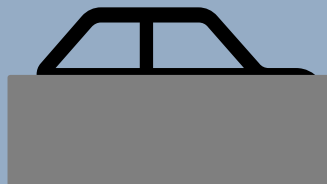


- ① Occlusion 이미지 10,883장 자동 추출
- ② 오검출 및 중복 이미지 제거 (2,179장)
- ③ 데이터 증강 (좌우반전, 확대/축소, 랜덤 변환 등)

→ 최종 약 4만 개 Occlusion 객체 이미지 학습

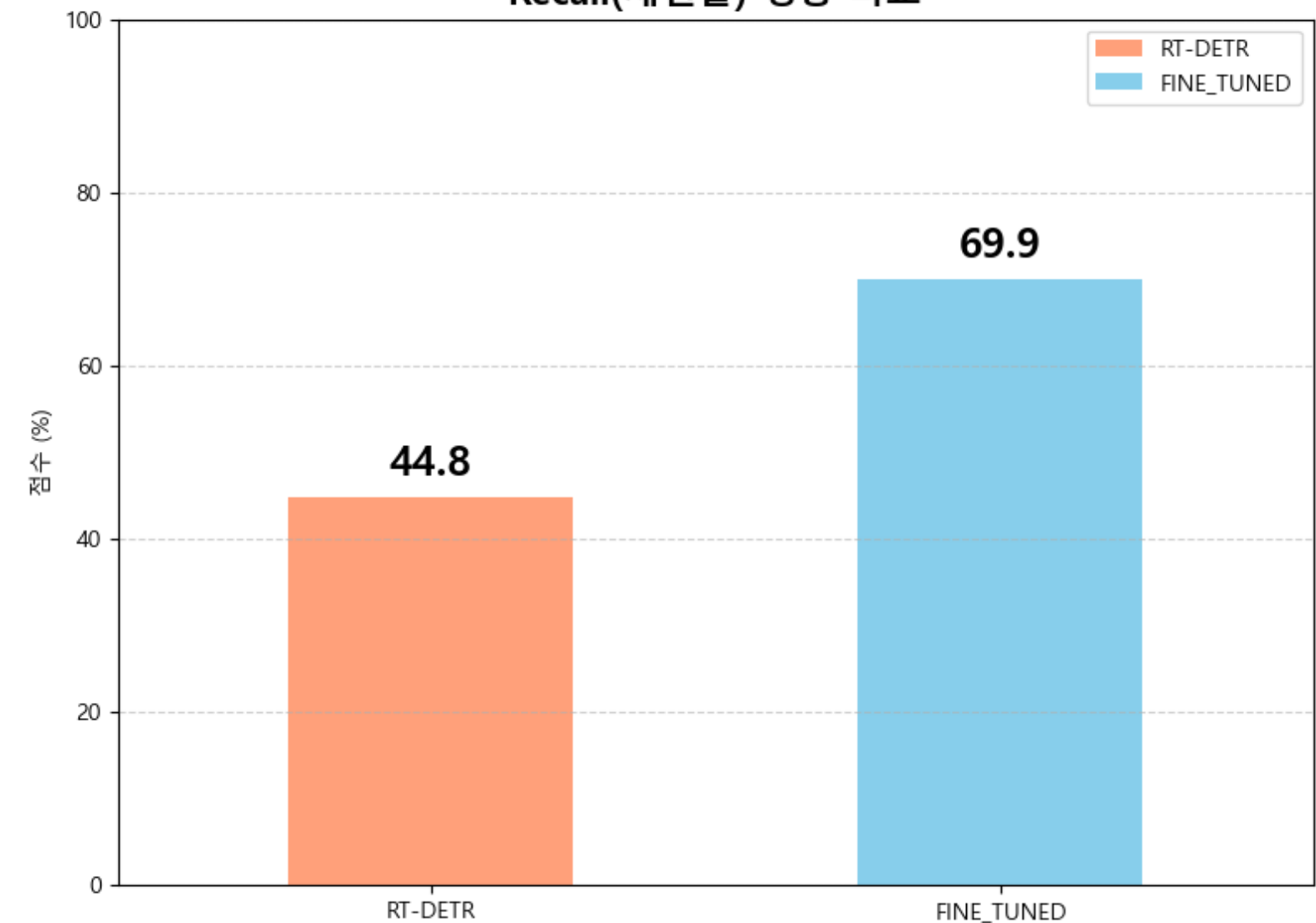
✓ 객체인식 - Baseline 모델과 비교

Confusion Matrix

	Positive	Negative
Occlusion-case	 정상 탐지	 미탐지
Normal-case	 오탐지	 정상 미탐지

✓ Recall(재현율) : $\text{정상 탐지 수} / (\text{정상 탐지 수} + \text{미탐지 수})$

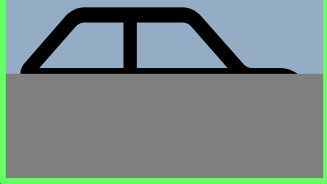
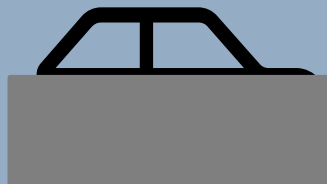
Recall(재현율) 성능 비교



놓치던 객체를 더 많이 검출

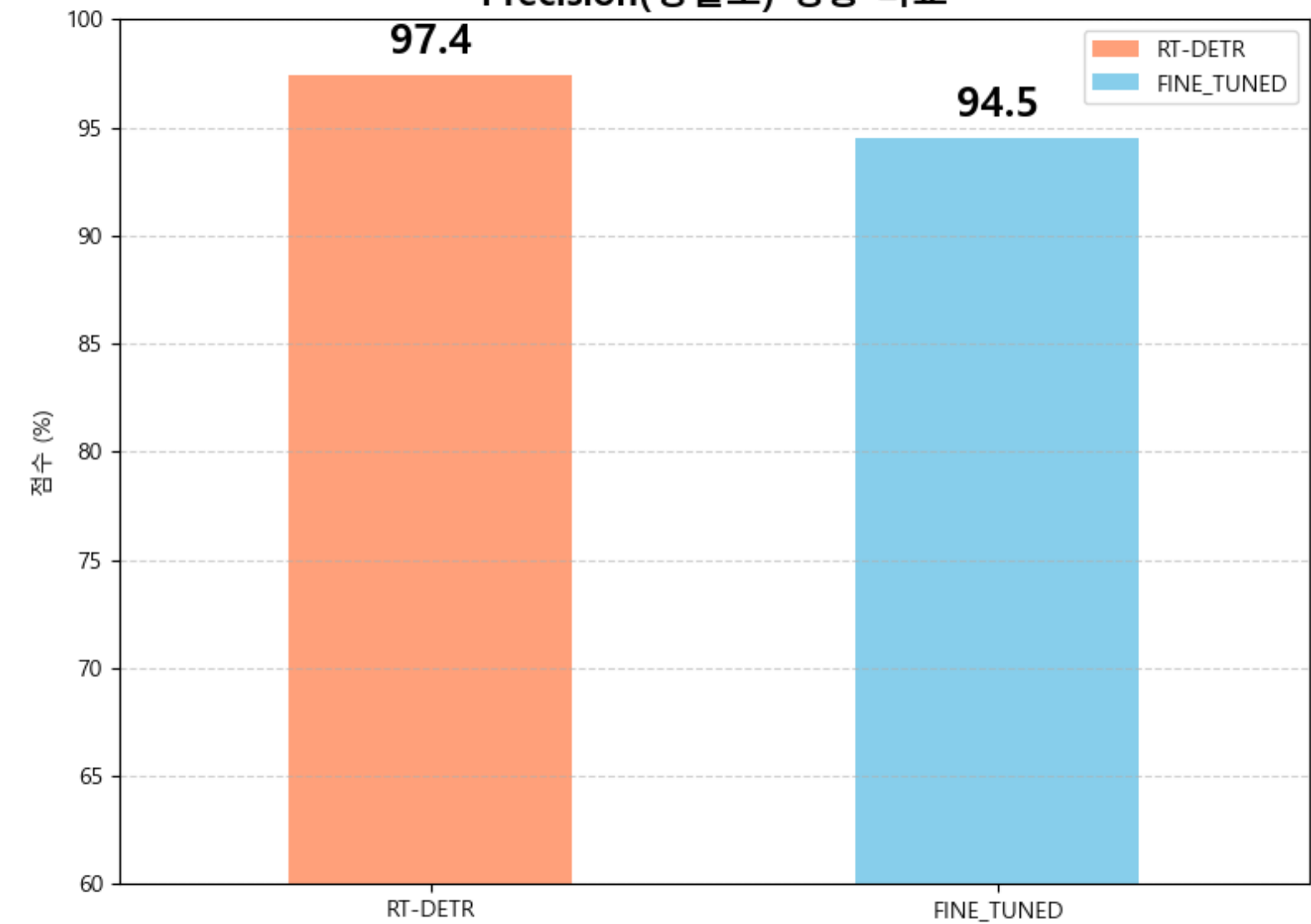
✓ 객체인식 - Baseline 모델과 비교

Confusion Matrix

	Positive	Negative
Occlusion-case	 정상 탐지	 미탐지
Normal-case	 오탐지	 정상 미탐지

✓ Precision(정밀도) : $\text{정상 탐지 수} / (\text{정상 탐지 수} + \text{오탐지 수})$

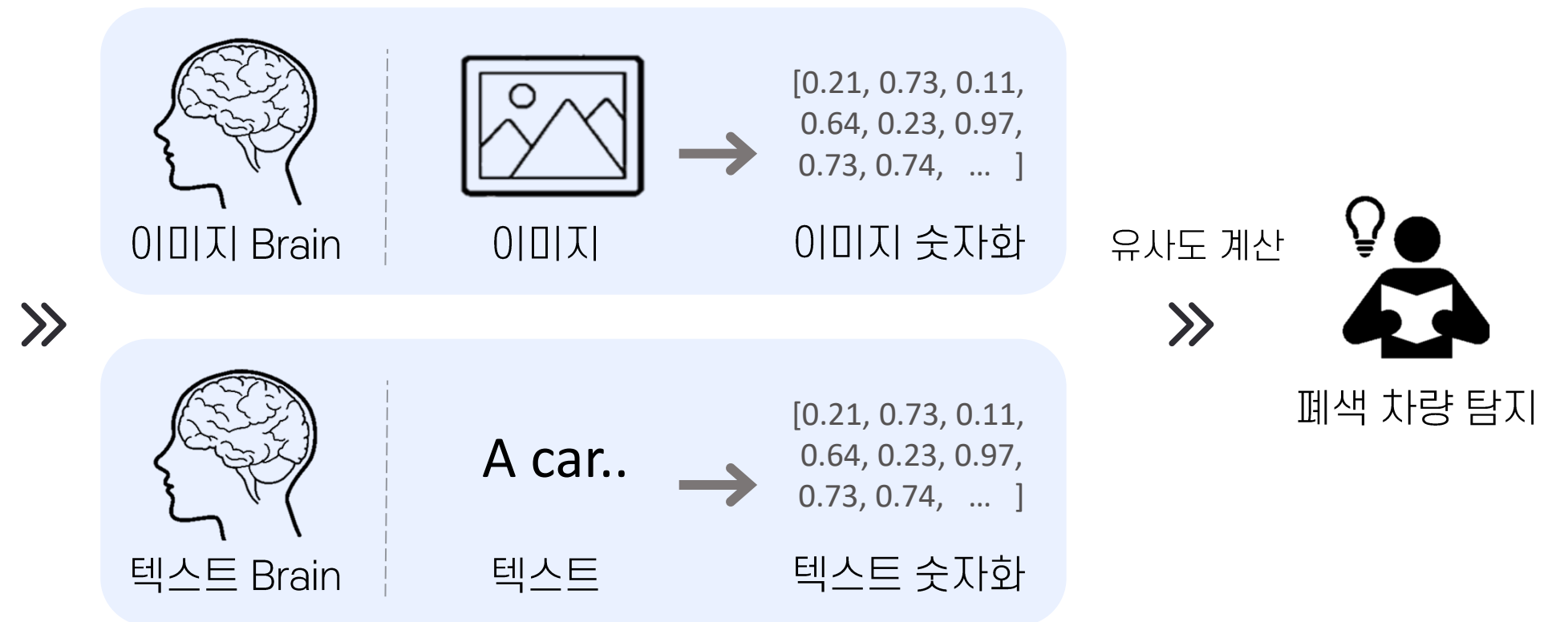
Precision(정밀도) 성능 비교



오탐지가 소폭 증가

✓ SigLIP 필터링

이미지와 텍스트의 의미를 연결하는 AI모델



✓ 기대 효과

EXPECTED EFFECT

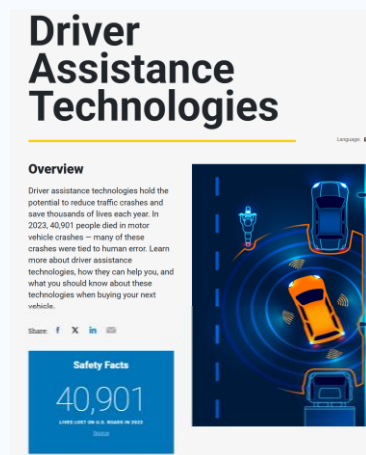
Occlusion 데이터 확보 자동화로 ADAS 인지 성능 개선



ADB 성능 향상

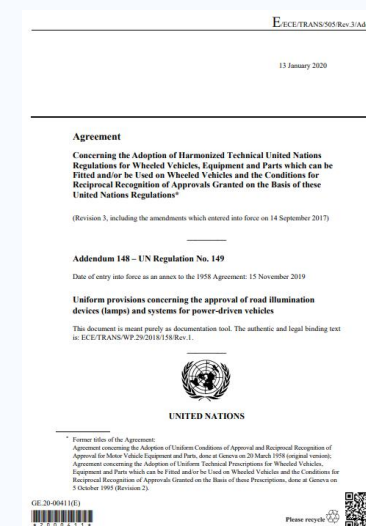


차량 안전 기능 강화



NHTSA (미국 고속도로 교통 안전청)

“조명 장치는 충분한 시야를 제공하면서도 다른 운전자에게 눈부심을 주지 않도록 설계되어야 한다.”



UNECE R149 (유엔 유럽경제위원회)

“운전자 보조 기술(ADAS)은 운전자와 탑승자 뿐만 아니라 다른 운전자와 보행자의 안전까지도 향상시키는 역할을 한다.”